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SNP marker detection and genotyping in tilapia

N. E. M. VAN BERS,* R. P. M. A. CROOIJMANS, M. A. M. GROENEN, B. W. DIBBITS and J. KOMEN
Animal Breeding and Genomics Centre, Wageningen University, PO Box 338, Wageningen, 6700 AH, The Netherlands

Abstract

We have generated a unique resource consisting of nearly 175 000 short contig sequences and 3569 SNP markers from the widely cultured GIFT (Genetically Improved Farmed Tilapia) strain of Nile tilapia (*Oreochromis niloticus*). In total, 384 SNPs were selected to monitor the wider applicability of the SNPs by genotyping tilapia individuals from different strains and different geographical locations. In all strains and species tested (*O. niloticus*, *O. aureus* and *O. mossambicus*), the genotyping assay was working for a similar number of SNPs (288–305 SNPs). The actual number of polymorphic SNPs was, as expected, highest for individuals from the GIFT population (255 SNPs). In the individuals from an Egyptian strain and in individuals caught in the wild in the basin of the river Volta, 197 and 163 SNPs were polymorphic, respectively. A pairwise calculation of Nei's genetic distance allowed the discrimination of the individual strains and species based on the genotypes determined with the SNP set. We expect that this set will be widely applicable for use in tilapia aquaculture, e.g. for pedigree reconstruction. In addition, this set is currently used for assaying the genetic diversity of native Nile tilapia in areas where tilapia is, or will be, introduced in aquaculture projects. This allows the tracing of escapees from aquaculture and the monitoring of effects of introgression and hybridization.

Keywords: next generation sequencing, NGS, *Oreochromis niloticus*, population genetics, SNP, Tilapia

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Introduction

Tilapia is the common name for a group of subtropical to tropical fresh water fish of the genera *Oreochromis*, *Tilapia* and *Sarotherodon*, belonging to the family of Cichlidae. Tilapias are one of the most widely farmed fish species in the world (FAO 2010), with *Oreochromis niloticus* being the most important tilapia species in aquaculture. *O. niloticus*, or Nile tilapia, is indigenous to Africa and the southwestern Middle East, but has been introduced to at least 87 countries (Froese & Pauly 2004), e.g. into many Asian countries for small-scale aquaculture (Gupta & Acosta 2004). The popularity of tilapia in aquaculture is the result of its fast growth rate, the year-round reproductive season, and its versatility to adapt to a large range of conditions which means that tilapia can be farmed in a wide range of farming systems (Coward & Little 2001; Canonico *et al.* 2005).

Some of the characteristics that make tilapias attractive as a culture species, also make them likely to, upon escape, flourish under feral conditions in the nations where they have been cultured or introduced to (Peterson *et al.* 2004; Canonico *et al.* 2005). Their introduction into native ecosystems, either on purpose or as escapees from aquaculture, poses a potential risk for the native biodiversity. This can be through predation or competition, but also through genetic effects such as hybridization, introgression and the decline in population size of the indigenous species, resulting in the loss of genetic diversity. On several occasions, an impact of cultured tilapia on native ecosystems has been reported (Twongo 1995; Goudswaard *et al.* 2002; D'Amato *et al.* 2007; Mwanja *et al.* 2010) (Canonico *et al.* 2005), e.g. a 80% reduction in native cichlids in Lake Nicaragua (McKaye *et al.* 1995). To monitor the effects of the introduction of invasive tilapias, it is important to have baseline data on what the level of (genetic) diversity of the native biota was before the introduction of the exotic tilapia. In addition, it is important to measure the level of introgression of farmed tilapia, and to trace the source of the escapees. As an alternative to introducing exotic tilapias, the aquaculture potential of native tilapia species should be exploited wherever possible, as native tilapia are believed to be a lesser threat to native biota than exotic tilapia (Canonico *et al.* 2005).

Genomic resources, such as DNA markers, are important tools for measuring genetic diversity and introgression and in estimating genetic parameters. The amount of genomic resources that are available for Nile tilapia are rapidly increasing, with the recent release of a draft

Correspondence: N. E. M. van Bers, Fax: +31317483929;
E-mail: nikkie.vanbers@gmail.com.
*Present address: NIOO-KNAW, Wageningen, The Netherlands.

genome sequence (http://www.broadinstitute.org), an extensive set of ESTs (Lee *et al.* 2010), two linkage maps (Kocher *et al.* 1998; Lee *et al.* 2005), a high resolution physical map (Guyon *et al.* 2012), a BAC-based physical map (Katagiri *et al.* 2005) and several sets of genetic markers e.g. (Lee & Kocher 1996; Kocher *et al.* 1998; Ezaz *et al.* 2004a; Lee *et al.* 2011). Genetic markers are important tools for selection purposes, e.g. in marker assisted selection and in association and QTL studies (Hulata 2001; Davey *et al.* 2011). Genetic markers also allow pedigree tracing and reconstruction, and the selection of animals with desired marker combinations. In addition, genetic markers are important tools in measuring the genetic diversity in and between populations. Until now, microsatellite and AFLP markers were the most commonly used markers in *O. niloticus,* and to the best of our knowledge no genome-wide set of SNP markers was publicly available for this species until now. SNPs are highly abundant in the genome, can be typed with a low error rate in high throughput assays and the genotyping results can easily be exchanged between laboratories or research groups (reviewed in Morin *et al.* 2004; Slate *et al.* 2009). Here, we describe the development of a set of 3569 SNP markers from twenty individuals of the widely cultured GIFT (Genetically Improved Farmed Tilapia) strain of *O. niloticus*. The GIFT strain is the result of a selective breeding programme which was initiated in 1988 by the WorldFish Center (WFC) and its research partners (Pullin *et al.* 1991; Eknath *et al.* 1993, 2007; Bentsen *et al.* 1998; Ponzoni *et al.* 2011). A subset of 384 of the SNPs was validated by genotyping individuals from the GIFT strain, and the wider applicability of the SNP set was proven by the typing of individuals from different strains and species of tilapia.

Methods

Sequencing and de-novo assembly

Genomic DNA was isolated from blood of 20 individuals of the 10th generation of the GIFT population of *Oreochromis niloticus* (Eknath *et al.* 1993; Gupta & Acosta 2004) using the Puregene system (Gentra, USA). To reduce the complexity of the data set, a reduced representation library (RRL) was created (Altshuler *et al.* 2000). For this, a pooled sample of DNA from the 20 individuals, containing a total amount of 100 μg of DNA, was digested with the *Rsa*I restriction enzyme (Fermentas, 200 U, in 1× tango buffer, o/n 37 °C), followed by dephosphorylation using CIAP (50 U), to avoid preferential adaptor ligation. The sample was size-fractionated on a 1% low-melting point agarose gel, and the fragments of 3000–3500 bp were isolated from the gel using β-agarase (NEB) treatment and were purified using phenol/sevag treatment and concentration by ethanol precipitation. The fraction of the genome covered by the RRL was estimated by dividing the DNA amount in the 3000–3500 bp agarose slice by the total amount of input DNA measured on the gel Doc XR (Biorad). The sequence library of the RRL was prepared using the Genomic DNA Sample Prep Kit (Illumina) according to the manufacturers' protocol, except for the omission of the phosphorylation step of the sample. Randomly sheared, adapter ligated fragments in the size-range of 170–250 bp were used as the starting material for sequencing on the Illumina Genome Analyzer (GA), resulting in 36 bp sequence reads.

The sequence data were filtered for quality as described in detail elsewhere (Van Bers *et al.* 2010). Briefly, the major filtering criterion was that a minimal quality score of 20 was required for all the nucleotides in a read in order for the read to be included in the *de-novo* RRL assembly. A quality score of at least 20 corresponds to a probability of >0.99 that the nucleotide is correctly called. The RRL assembly was performed using the program SSAKE (default settings) (Warren *et al.* 2007). All assembled sequences of at least 37 bp are further referred to as contigs.

SNP detection, validation, selection and genotyping

Single nucleotide polymorphisms (SNPs) were detected by mapping quality filtered reads on the *de-novo* RRL assembly. The sequence reads were filtered for quality prior to mapping (Van Bers *et al.* 2010). In brief, a minimal quality score of 10 was required for all the nucleotides in a read in order for the read to be included in the mapping, and reads that were overabundant (observed more than five times the expected sequence depth of 34) were excluded from the analysis as they are potentially derived from repetitive sequences. The expected sequence depth is derived from the number of nucleotides sequenced (33 million reads × 36 nt) divided by the estimated fraction of the genome sequenced (3.5% of 1 billion basepairs).

All the quality filtered reads were used for mapping onto the *de-novo* assembled contig sequences using MAQ (version 0.6.6, default settings) (Li *et al.* 2008). Detected variation was classified as a SNP when the following requirements were met: (i) the minor allele was observed at least three times; (ii) the best mapping read had a mapping quality (Q) of at least 40; and (iii) the consensus quality (C) was at least 30. Both mapping and consensus quality are phred like quality scores (Li *et al.* 2008). The mapping quality measures the probability that the true alignment is not the one that is found. A mapping quality score of 40 corresponds to a 10^{-4} chance that the alignment found is not the true position for the read. MAQ produces a consensus genotype from the aligned reads, and a consensus quality of 30 corresponds to a probability of 10^{-3} that the consensus genotype is incorrect.

A design quality score (which can range between 0 and 1) was assigned to the SNPs using the online 'assay design tool' on the Illumina website (http://www.illumina.com). As a first validation of the detected SNPs, a random set of 20 contigs larger than 500 bp were re-sequenced. For this, PCR primers were designed using the Primer3 software (Rozen & Skaletsky 2000), and amplicons were sequenced on the ABI3730 using the ABI sequencing big dye kit (Applied Biosystems).

Single nucleotide polymorphism genotyping was performed in a 384 SNP multiplex assay using the GoldenGate Assay (Illumina). For a SNP to be included for genotyping an illumina design score of at least 0.65 was required. However, preference was given to SNPs with a design score of 0.75 or more, with an observed read depth at the SNP position of 15–35 reads. Not more than one SNP was selected per contig. All the 20 SNPs used for validation by PCR were included in the genotyped SNP set.

Oligonucleotides flanking a SNP were designed according to Illumina's design program specifications, and the assay was deployed on a BeadXpress platform using the Veracode technology. All SNPs that were found to be polymorphic in our genotyping analysis have been submitted to dbSNP with accession numbers ss244316446–ss244316740. Information on the performance of the SNPs (e.g. the call rate and MAF) can be found in File S1.

Origin of the samples

The genotyping was performed on DNA from two sets of 20 individuals of the 10th generation of the GIFT population (Eknath *et al.* 1993; Gupta & Acosta 2004) which were genotyped in separate experiments. The GIFT fish originated from the WorldFish Center (WFC) in Malaysia, and all 20 individuals used for the detection of the SNPs were included in the genotyping assay. An additional 30 *O. niloticus* individuals collected from the wild at six different geographical locations in the basin of the river Volta were genotyped. These locations are: Ghana (eight individuals), Burkina Faso (six individuals), Mali (six individuals), the Republic of Benin (four individuals), the Togolese Republic (two individuals) and Ivory Coast (four individuals). The Egyptian strain of *O. niloticus* included for genotyping (21 individuals) originated from the WFC in Abbassa (Fessehaye *et al.* 2009). The grandparents of these individuals were produced in 2000 from all possible diallelic crosses between four Egyptian strains of *O. niloticus,* originally collected from: (i) the Ismailia Canal East of the Nile Delta near Abbassa in 1998; (ii) a hatchery near Lake Maryout, a lake in the northwestern part of Egypt south of Alexandria in 1995; (iii) the El-Zawia region North of the Nile Delta near Lake Burullus in 1995; and (iv) a wild population from the Aswan region (Rezk *et al.* 2002; Charo-Karisa *et al.* 2005). The 20 *Oreochromis aureus* individuals genotyped are from a laboratory population descending from fish originally collected from Lake Manzala (Egypt) approximately 30 years ago (D. Penman, personal communication). Also the 20 *Oreochromis mossambicus* individuals genotyped are from a laboratory population, descending from fish originally collected approximately 30 years ago from the lower Zambezi River in Zimbabwe (D. Penman, personal communication). The single individual that was genotyped from the isogenic line of *O. niloticus* was also used for the whole genome sequencing project by the BROAD institute. This line was developed from a population from lake Manzala in Egypt (Sodsuk & McAndrew 1991). This isogenic line was initially developed by mitotic gynogenesis, and was maintained by sex reversal and crossing (Sarder *et al.* 1999; Ezaz *et al.* 2004b).

Sequence alignments

The unpublished tilapia whole genome assembly was downloaded from http://www.broadinstitute.org/ftp/pub/assemblies/fish/tilapia/Orenil1/ on 4 February 2011. The other fish genomes used for sequence alignment were downloaded from the UCSC genome browser. These genomes were: (i) the genome assembly GasAcu1 of *Gasterosteus aculeatus*; (ii) danRer6 of *Danio rerio*; (iii) OryLat2 of *Oryzias latipes*; and (iv) v4.0 of *Takifugu rubripes* (JGI4.0/fr2). *Oreochromis niloticus* ESTs were downloaded from the NCBI EST database on 27 January 2012. All alignments were performed using the LASTZ software (Harris 2007), using the default parameters, except for coverage = 80, identity = 60, C = 2 and ydrop is 3400. The adjustments in the parameter settings were primarily done to account for the small size of the query sequences.

Data analysis

The Genome studio software (Illumina) was used for the analysis of the genotyping data. The Genetic Distance Matrix program version 3.69 (PHYLIP) was used to compute Nei's genetic distance (Nei 1972) from the genotypes of the 109 successfully typed individuals. The 20 animals of the discovery panel and the three animals that failed in the genotyping analyses were not used for the calculation of Nei's genetic distance. The genetic distances were used to draw a Neighbour joining tree with the MEGA5 software (Tamura *et al.* 2011).

Results

Sequencing and sequence assembly

To generate a set of SNPs for tilapia, we sequenced a reduced representation library (RRL) derived from a pooled DNA sample of 20 individuals. The RRL represented an estimated 3.5% of the genome. In total, 33 million short sequence reads (36 nucleotides) were generated. At the time of the analysis no publicly accessible reference genome was available for tilapia, so a *de-novo* RRL assembly was performed of sequence reads filtered for quality. After filtering for quality, 9.1 million reads were retained for use in the assembly. This assembly resulted in 174 950 contigs with a N50 of 73 nucleotides. The N50 length of an assembly is the length x, such that 50% of the total assembled sequence is contained in segments of length x or greater (Adams *et al.* 2003). The total length of the assembled RRL contigs was 13 million basepairs. This correspond to ~1.2% of the tilapia genome which has an estimated size of 1060 Mbp (Gregory 2011).

Recently, a draft whole genome sequence of tilapia (Orenil1) has become publicly available, which is currently being assembled by the Broad Institute (Cambridge, USA). Sequence alignment of the 174 950 generated contigs with this draft tilapia genome sequence, shows that 90% of the *de-novo* assembled RRL contigs align to the whole genome sequence of tilapia (Table 1). In contrast, only 3–4% of the RRL contigs align to other publicly available fish genomes (Table 1), for example, stickleback (*Gasterosteus aculeatus*), zebra fish (*Danio rerio*), pufferfish (*Takifugu rubripes*) and medaka (*Oryzias latipes*).

SNP detection and selection

The *de-novo* assembled RRL contigs were used as the reference for the mapping of nearly 22 million sequence reads filtered for quality. This number of reads is higher than the number of sequence reads used for the assembly, which is due to the more stringent filtering criteria applied to the reads used for the assembly (e.g. a higher minimum quality score was required for all the nucleotides in the assembly). This was done because sequencing errors have a higher impact on the assembly than on the SNP detection, for example, they may lead to smaller and putatively erroneous contigs. During SNP detection, the requirement that a SNP needs to have been observed multiple times provides additional quality filtering. Mapping of the sequence reads allowed the detection of 3569 SNPs, with an average read depth of 21 at the SNP position. The SNPs are distributed over 3246 contigs, of which 3012 (93%) could be aligned onto the draft tilapia genome sequence (Table 1). As a first validation of the detected SNPs, a subset of 20 SNPs was selected for amplification by PCR and sequencing. Amplification was successful for 18 of the selected fragments, and 100% of the SNPs that were predicted in the selected fragments were confirmed to be polymorphic by sequencing.

The minor allele frequency (MAF) of a SNP can be estimated from the relative number of times that the minor allele was observed in the sequence reads covering the SNP position. The average estimated MAF of all 3569 SNPs was 0.34 (±0.09) in the pool of individuals used for sequencing. This relatively high number reflects the selection for relatively common SNPs inflicted by the requirement that for SNP calling the minor allele needs to have been observed at least three times.

For 115 SNPs, three different alleles were observed, which rendered them unsuitable for use in our subsequent genotyping analysis. A design score was assigned to the remaining 3454 SNPs for which two alleles were observed. A design score of a SNP predicts the suitability of the SNP for genotyping, taking into consideration, for example, the length and complexity of the sequence flanking the SNP, and this score typically ranges from 0 to 1 when calculated with the Assay Design Tool (Illumina). For 1856 SNPs the design score was higher than 0.65, which is sufficiently high to consider inclusion in the assay design. Out of the 1856 SNPs, 384 were selected for subsequent genotyping. These SNPs were located on 384 different RRL contigs, and 370 (96%) of these contigs could be successfully aligned to the draft tilapia genome

Table 1 Alignment of the contigs to the genomes of different fish species

	Database used for alignment	No. of nucleotides in database (Mbp)	Hits of total contig-set (174 950 contigs, 13 Mbp)	Hits of SNP-containing contigs (3246 contigs, 535 078 bp)	Hits of contigs of 384 SNP set
Oreochromis niloticus	Genome assembly	924	157 010 (90%)	3012 (93%)	370 (96%)
Gasterosteus aculeatus (stickleback)	Genome assembly (gasAcu1)	463	7590 (4%)	270 (8%)	73 (19%)
Danio rerio (zebra fish)	Genome assembly (danRer6)	1 500	4399 (3%)	139 (4%)	21 (5%)
Oryzias latipes (medaka)	Genome assembly (oryLat2)	886	6071 (3%)	251 (8%)	52 (14%)
Takifugu rubripes (fugu)	Genome assembly	400	4376 (3%)	162 (5%)	43 (11%)

sequence (Table 1). Using the same methods, 73 contigs were aligned to the stickleback genome, 21 to the zebra fish, 43 to the pufferfish and 52 to the medaka genome (Fig. 1 and Table 1). Two contigs aligned to the genomes of all five species, and 21 contigs aligned to the genomes of four of the species analysed. In addition, 5 of the 384 contig sequences aligned to *Oreochromis niloticus* EST sequences (see Table S1).

Genotyping

The 384 SNP set was first used for the genotyping of the 20 individuals from the 10th generation of the GIFT strain that were used as the discovery panel for the detection of the SNPs. All the animals were successfully typed (call rate >0.8), with an average call rate of 0.94. In total, 293 SNPs could be typed in at least 75% of the animals. Of these SNPs, 80% were polymorphic, and 59 SNPs were not polymorphic. The fact that some of the non-polymorphic SNPs were not successfully typed in all the animals of the discovery panel rendered the possibility that a subset of these SNPs was polymorphic in an animal that failed in the genotyping. However, 53 SNPs turned out to be fully homozygous for the same allele in all the animals of the discovery panel. The 293 SNPs that were successfully typed in at least 75% of the animals had an average MAF of 0.24 (±0.16). This number increased to 0.30 (±0.12) when only the SNPs that were truly polymorphic were considered. The observed MAF and call frequency for all the SNPs can be found in File S1.

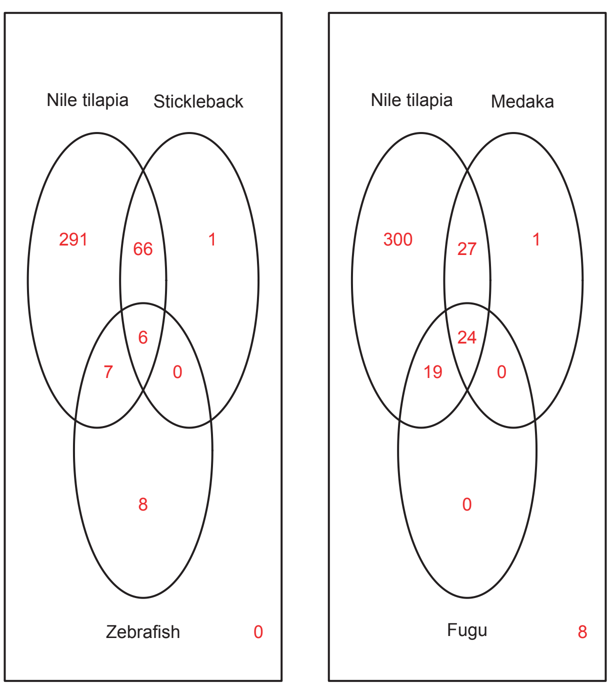

Fig. 1 Venn diagrams showing the number of SNP-containing contigs (out of the total number of 379) for which a mapping position was found in one or more model fish species with a sequenced genome. The number in the right bottom corner of each of the diagrams indicates the number of contigs (from the total of 379) that are not covered in the respective diagram.

Subsequently, we used the 384 SNP set for genotyping tilapia individuals from different species, strains and different geographical locations to investigate the wider applicability of the developed SNP set. The observed MAF and call frequency of the SNPs is provided for each of the strains in File S1. In total, 112 individuals were analysed, of which 109 samples had a call rate that was sufficiently high (>0.9) for an analysis of genetic relatedness (Table 2). The three excluded samples comprised of two samples from the Ivory Coast (Volta basin), and one *O. aureus* sample. Based on the genotypes, we calculated Nei's genetic distance between all the animals, and used the distances to infer phylogenetic relationships between the individuals (Fig. 2). This shows that the animals of different species and (to some extent) geographical origin can be distinguished based on their genotypes determined with the 384 SNP set. As expected, the average heterozygosity was highest for the individuals from the 10th generation of the GIFT strain, and this heterozygosity (0.31) was only slightly lower than the one observed for the animals that were used as the discovery panel (0.34) (Table 2). A lower heterozygosity of 0.2 was found for the Egyptian strain of *O. niloticus*. The samples collected in the basin of the Volta river had a heterozygosity of 0.13, with 156 SNPs giving informative genotypes for more than 75% of the individuals. This allows the discrimination of these samples from the other strains of *O. niloticus* used in this study (Fig. 2). In addition, the samples from the different geographical locations do often, but not always, cluster together in the tree which is expected to enable to a limited extent the assignment of the individuals to an exact geographical location (Fig. 2).

The whole genome sequence of tilapia is derived from a single individual from an isogenic, fully homozygous line. We included another individual (O_nil_seq) from the same clonal line in our genotyping analysis. The isogenic character of the line is reflected in the fact that 98% of the SNPs were homozygous in this individual (Table 2). The isogenic clonal line was originally developed from an Egyptian strain of *O. niloticus* (Sarder *et al.* 1999; Ezaz *et al.* 2004b)). In our genetic distance analysis, O_nil_seq is placed between the GIFT strain and the Egyptian strain of *O. niloticus* (Fig. 2).

Discussion

At the onset of our study, no sequenced reference genome was available for *O. niloticus.* To allow SNP

Table 2 Genotyping of different strains of tilapia

Origin	No. of samples	Average heterozygosity	Polymorphic SNPs*	No. of SNPs successfully typed in >75% of samples
Oreochromis niloticus, GIFT, discovery panel	20	0.34 ± 0.22	234 (256)	293
O. niloticus Volta basin	28	0.13 ± 0.18	156 (163)	301
O. niloticus clonal line	1	0.024	7	297
O. niloticus Egypt	21	0.20 ± 0.22	190 (197)	305
O. niloticus GIFT	20	0.31 ± 0.22	247 (255)	302
O. aureus	19	0.04 ± 0.17	21 (30)	298
O. mossambicus	20	0.04 ± 0.16	26 (31)	288

*This is the number of SNPs that were polymorphic and were successfully typed in 75% of the individuals of the respective population. The number given in brackets is the total number of SNPs that was found to be polymorphic in the respective population.

detection, we performed a *de-novo* assembly of short sequence reads generated from a Reduced Representation Library (RRL) of a pooled DNA sample of 20 fish from the GIFT strain of Nile tilapia. The main advantages of using a RRL of a pool of individuals for SNP discovery are: (i) that the minor allele frequency of the detected SNPs can be estimated from the read depth at the SNP position; and (ii) that the SNPs should be more or less evenly distributed over the entire genome even though the complexity of the data set is reduced (Altshuler *et al.* 2000; Van Tassell *et al.* 2008; Van Bers *et al.* 2010, 2012; Davey *et al.* 2011). The assembly of the short sequence reads resulted in nearly 175 000 contig sequences, with a total length of 13 million nucleotides. Recently, the first build of the tilapia whole genome sequence, generated by the BROAD institute, became publicly available. This allowed the positioning of 90% of the generated RRL contigs onto the draft genome. The remaining 10% of RRL contigs that was not found back on the draft genome sequence may represent a part of the genome that is not present in the current first assembly of the genome. However, we cannot exclude that 10% of the RRL contigs do not align due to a high divergence between the genome sequence and the RRL contigs, which hampers the alignment of the contigs.

The current availability of a reference genome for Nile tilapia bypasses the need for a *de-novo* assembly for SNP detection. This would allow for the detection of a relatively larger number of SNPs, because not all of the sequenced data are incorporated in a *de-novo* assembly. Our RRL assembly covered less than 1.3% of the genome, while the data set was estimated to represent 3.5% of the genome. In addition, also the number of assayable SNPs increases as a result of the availability of a genome sequence. This is because a whole genome sequence provides sufficient flanking sequence information for nearly all the SNPs, which results in an increase in the number of SNPs suitable for inclusion in a genotyping assay. Furthermore, the use of a reference genome allows the selection of SNPs in non-repetitive regions, which are more likely to result in successful genotyping. In our analysis, 53 SNPs (14%) turned out to be fully homozygous in all the animals used in the discovery panel. This may be due to: (i) additional SNPs flanking the SNP, which results in the selective amplification of only one of the two alleles; (ii) sequencing errors (although our requirement of observing the minor allele at least three times makes this less likely); and (iii) repetitive regions that have collapsed into one contig sequence, which makes that putative SNPs are detected that actually locate to different regions in the genome. Using the same methods in other species (Kerstens *et al.* 2009; Van Bers *et al.* 2012) we did not observe such high numbers of non-polymorphic SNPs, and high number of SNPs failing during genotyping, which suggests that this may be specific for this data set or this species. In contrast, also in Atlantic salmon high numbers of putative SNPs did not result in reliable genotypes (Glover *et al.* 2010; Karlsson *et al.* 2011).

Mapping of the sequence reads onto the *de-novo* assembled RRL contigs, allowed for the discovery of 3569 novel SNPs, located on 3246 contigs. From these SNP-containing contigs, 93% could be placed onto the whole genome sequence of *O. niloticus*, and 96% of the contigs harbouring the 384 genotyped SNPs could be mapped. This slightly larger number is most likely reflecting that, due to the larger flanking sequence, the SNPs in larger contigs have a higher chance of being suitable for inclusion in a genotyping assay and these larger contigs also have a relatively higher chance of finding sufficient sequence similarity on the whole genome sequence. In total, 1856 SNPs passed the criteria for inclusion in an Illumina genotyping assay, and even though we have given preference to the best scoring SNPs in our 384 SNP set, we expect that the majority of the remaining assayable SNPs will also be informative after genotyping analysis.

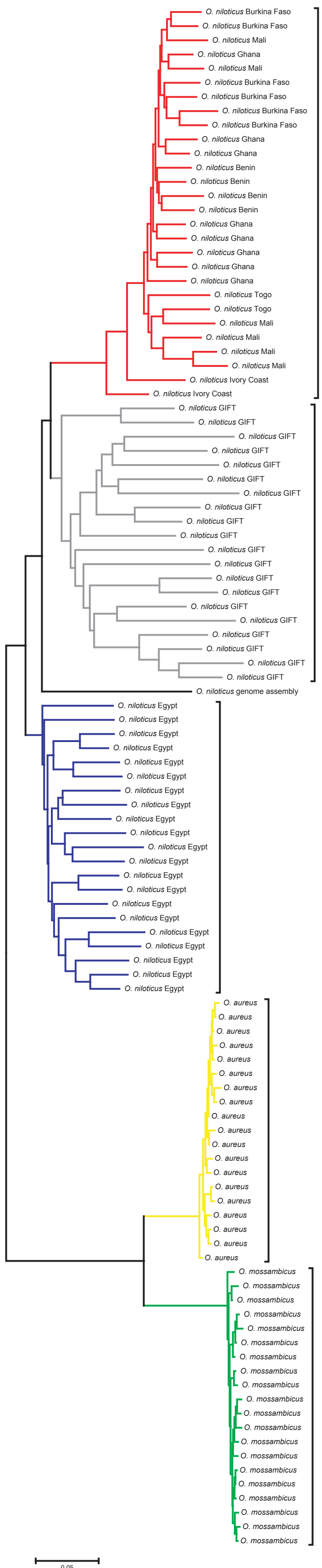

Fig. 2 The phylogenetic tree is based on Nei's genetic distances calculated from pairwise comparison of the genotypes (261 SNPs) of the 109 individuals. The branch colours identify all the different strains and species of Tilapia used in this analysis. The tree shown was constructed using the Neighbour Joining method.

To assign a more exact chromosomal position to the 384 SNPs of our set, 272 SNPs were included on a Radiation Hybrid (RH) physical map of Nile tilapia (Guyon *et al*. 2012). This map was constructed with in total 1296 markers, e.g. gene, BAC, microsatellite and SNP markers. The markers were assigned to 81 RH groups, and the inclusion of microsatellites that were also used for the construction of the two genetic maps of tilapia (Kocher *et al*. 1998; Lee *et al*. 2005) enabled the assignment of the RH groups to all 24 linkage groups, which correspond to 22 tilapia chromosomes (Majumdar & McAndrew 1986). Of the 272 SNPs included on the map, 241 were grouped into RH groups. Of these SNPs, 196 could be positioned on linkage groups corresponding to 21 out of the 22 tilapia chromosomes. Only LG10, which corresponds to the smallest tilapia chromosome, does not contain any of the SNP markers included in the assay (Guyon *et al*. 2012). The data from the RH map indicate that the SNPs are genome-wide distributed.

A putative mapping position on the genomes of one or more of the other fish species with a sequenced genome could be assigned to a subset of the 384 genotyped SNPs (5–19%). This low number reflects the low sequence similarity due to the relatively long divergence time between tilapia and other model fish species with a sequenced genome. The availability of a mapping position on other (annotated) genomes for some of the SNPs may facilitate the identification of candidate genes underlying QTL regions. Interestingly, of the 14 contigs that did not find a mapping position on the tilapia genome, 8 could be mapped onto the zebra fish genome, and 1 was mapped to the stickleback genome. The majority of these, for example, seven of the SNPs sequences found in zebra fish, and the single SNP sequence found in stickleback, were located in regions that could not be assigned to a specific chromosome (Chr unassigned). This indicates that the SNP sequences are located in regions that are problematic in the genome assembly, which may explain why these SNPs could not be placed on the Tilapia genome yet.

The use of a pool of individuals for SNP discovery allows the a priori selection of SNPs with a relatively high MAF for genotyping analysis. This increases the chance of a SNP being relatively ancient and therefore being polymorphic in populations or strains other than the one in which they were originally discovered. The multiple strain origin of the GIFT population further optimized the chance of the SNPs being informative in other Nile tilapia strains. The GIFT strain is a synthetic strain, composed of eight different strains of Nile tilapia: four wild strains from Africa and four strains cultured in Asia. The wild strains were collected in Egypt, Ghana, Kenya and Senegal (Eknath *et al.* 1993), and the farmed strains are regarded as having originated from Egypt and Ghana (Pullin & Capili 1988; Macaranas *et al.* 1995). The developed SNP set was indeed performing well in the other strains of Nile tilapia, with the highest level of heterozygosity in the Egyptian strain which has a partially overlapping genetic background with the GIFT strain. The phylogenetic tree derived from the genetic distance analysis of the genotypes (Fig. 2) places the GIFT strain in between the Egyptian strain and the African *O. niloticus* samples, which reflects the origin of the GIFT strain. We also typed the 384 SNPs on two other tilapia species, *O. aureus* and *O. mossambicus*. Nearly all SNPs (99% and 95% respectively) that could be typed in the GIFT population were also successfully typed in these two species, which shows a relatively high level of sequence conservation between these three tilapia species. In contrast, the actual polymorphisms were not conserved between the species with only 10% of the SNPs being polymorphic in *O.aureus* and *O. mossambicus*, which reflects the fact that the SNPs were originally developed in *O. niloticus*. Besides the evolutionary distance, this high number may also reflect the relatively low population size at which the genotyped individuals have been kept for over 20 years (D. Penman, personal communication), and the actual number of SNPs polymorphic in other (wild) stocks of *O. mossambicus* and *O. aureus* may in fact be higher. One individual fish (O_nil_seq) derived from the same clonal line as was used for the whole genome sequencing project of tilapia was also included for genotyping with the 384 SNP set. This individual was expected to be fully homozygous, but 7 SNPs turned out to be polymorphic. Manual inspection of the genotyping profiles of these SNPs indicate that most of these SNPs showed ambiguous clustering of the three genotype classes (AA, AB and BB). This indicates that the obtained 2.4% heterozygosity of O_nil_seq is most likely to be overestimated, due to genotyping errors.

The SNP set presented here performed well in all stocks tested, and allowed the discrimination of different strains and species of tilapia. The discrimination of the geographical origin of the samples from the basin of the river Volta was only possible to a limited extent (Fig. 2), which is presumably due to the relatively low number of samples included for each of the regions (see also Morin *et al.* 2009). SNP markers have been shown to be the markers of choice for population, individual and parental assignment in e.g. multiple species of Salmon (Smith *et al.* 2007; Smith & Seeb 2008; Habicht *et al.* 2010; Hauser *et al.* 2011; Seeb *et al.* 2011). In addition, SNPs are used to determine the source origin of animals escaped from aquaculture (Glover *et al.* 2008; Beacham *et al.* 2011; Hess *et al.* 2011). Depending on the question posed, the decision on which number and which SNP to use may vary (Morin *et al.* 2009). For kinship and individual identification, SNPs with higher MAF may result in a higher power (Chakraborty *et al.* 1999; Krawczak 1999). However, for population structure studies the preferential selection of SNPs with a high MAF in the population where they were originally identified increases the risk of creating ascertainment bias (Wakeley *et al.* 2001; Brumfield *et al.* 2003; Morin *et al.* 2004). The data set presented here is developed from DNA of individuals from the GIFT strain, and randomly selecting SNPs from this data set for population genetic studies poses the risk of ascertainment bias. To help in the selection of SNPs, we have presented information on the call frequency and MAF of the SNPs in all the stocks analysed. This will allow the selection of subsets of SNPs with characteristics depending on the population and the desired application. For example, we have used a subset of 150 SNPs with a relatively high MAF for parental assignment in the GIFT population (T. Q. Trịnh, N. E. M. van Bers, J. Komen, submitted). The high MAF resulted in a high discriminative power, which allowed us to distinguish between closely related animals.

Africa has a huge potential for tilapia farming. However, an expert consultation in 2002 cautioned against the introduction of GIFT fish to Africa due to concerns regarding the contamination of the native biodiversity by escapees from aquaculture (Gupta *et al.* 2004). To overcome this, the GIFT methodology is being applied to genetically improve indigenous Nile tilapia in several countries in Africa (e.g. in Egypt, Cote d'Ivoire and Ghana) (Gupta *et al.* 2004). However, there remains the importance of monitoring the occurrence and effects of escapees from aquacultures. We are currently using a subset of 192 of the SNPs described here to investigate the baseline genetic diversity of native Nile tilapia in the basin of the river Volta in Western Africa.

In conclusion, we have described the development of a 384 SNP set from the widely cultured GIFT strain of Nile tilapia. We show that this set allows the distinction between individuals from different strains and species of Tilapia. This, together with the genome-wide distribution of the SNP markers, will make this a valuable set for future use in tilapia breeding, as well as for conservational purposes by creating essential understanding of the native biodiversity of regions where tilapia is, or will be introduced in aquaculture. This is important for the efficient monitoring of the future effects of aquaculture on native biodiversity.

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NEMvB performed the assembly, SNP discovery, SNP selection, sequence alignments, analysed the genotyping data and drafted the manuscript. BWD and RPMAC prepared and genotyped the tilapia samples. NEMvB, RPMAC, JK and MAMG participated in the design of the analysis. JK, RPMAC and MAMG all provided comments on the manuscript. All authors read and approved the final manuscript.

Data accessibility

All SNPs that were found to be polymorphic in our genotyping analysis have been submitted to dbSNP with accession numbers ss244316446–ss244316740. The short reads have been submitted to the Short Read Archive (NCBI) in SRA049722. The contigs are available from the authors upon request. The observed MAF and call frequency for all the SNPs in each of the strains tested can be found in File S1.

Supporting information

Additional supporting information may be found in the online version of this article.

File S1 The accession numbers and map positions of all the SNPs, and the observed MAF and call frequency for all the SNPs in each of the strains tested.

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